Use Cases

for

Crowdy

Version 3.2

Prepared by Tan Leong Jun Joseph, Teo Kok Wei Oscar, Mao Yikai, Xu Hanwen, K Eswaran

SCSJ Group 3, SC2006 Software Engineering, Nanyang Technological University

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Revision History

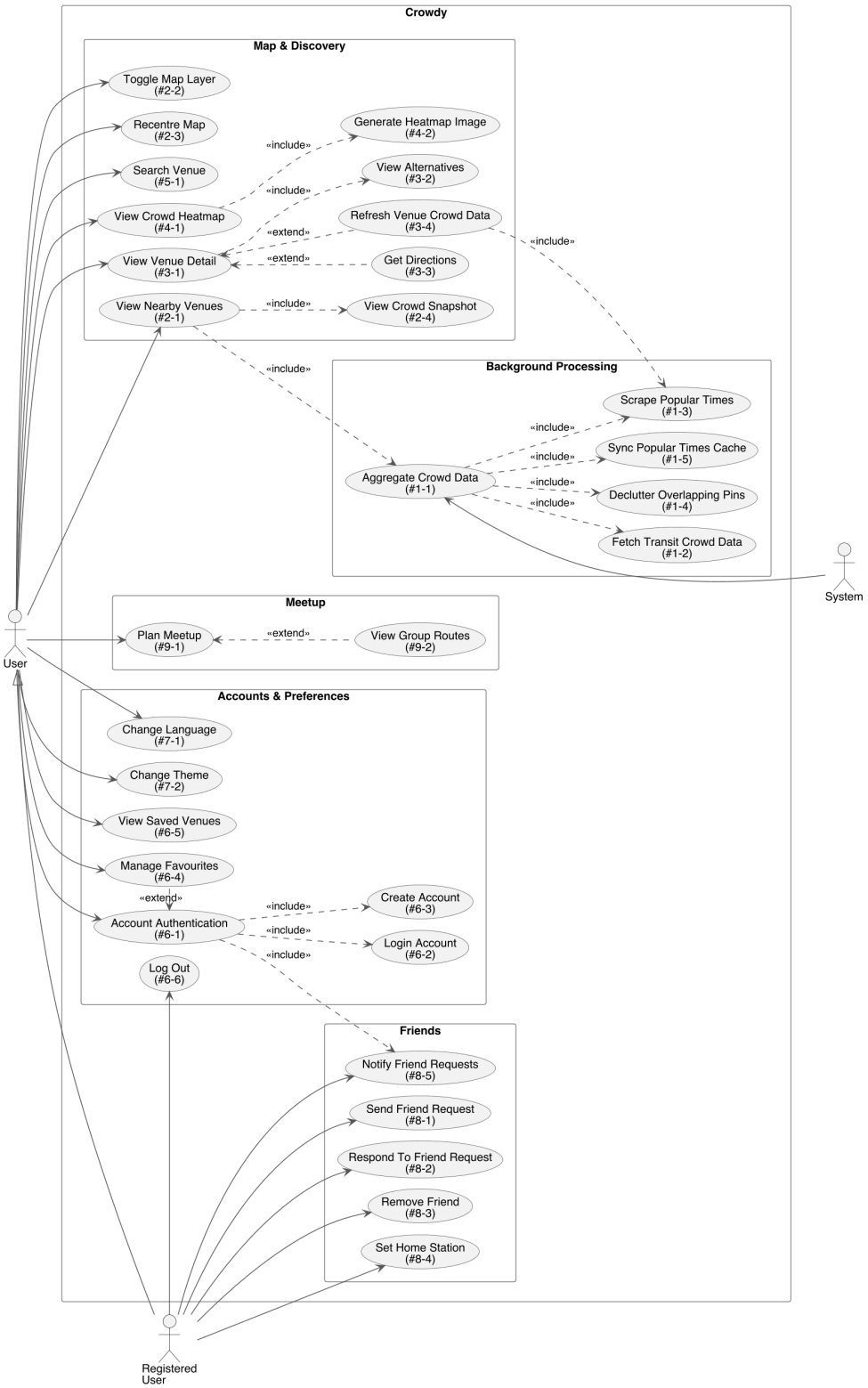
Name	Date	Reason For Changes	Version
SCSJ Group 3	16/09/2026	Initial use case model (Lab 1)	1.0
SCSJ Group 3	21/10/2026	Complete use case descriptions (Lab 3)	2.0
SCSJ Group 3	01/11/2026	Final use case model for the delivered app (Lab 5)	3.0
SCSJ Group 3	06/11/2026	Popular Times scraping pace and start-up rescrape prompt	3.1
SCSJ Group 3	08/11/2026	Start-up view: the whole of Singapore with the heatmap on and both layers off	3.2

1. Use Case Diagram

Figure 1 shows Crowdy's complete use case model. The descriptions that follow are grouped by the functional requirement each use case realises. Other use cases are referred to by name and ID, for example SyncPopularTimesCache (#1-5).

Crowdy has three actors. User is anyone using the app, with or without an account. Registered User is a User who has signed in, and can do everything a User can, plus the use cases that need an account. System is Crowdy's backend acting on its own schedule.

Each «include» or «extend» label sits above its dashed line, or beside it where the line runs up and down. The arrow points to the use case being included or extended. A full-resolution copy of Figure 1 is in SCSJ_Group3_Lab5_Diagrams/usecase-diagram.png.



Each «include» or «extend» label sits above its dashed line, or beside it where the line runs up and down. The arrow points to the use case being included or extended.

Figure 1. Crowdy use case diagram

2. Use Case Descriptions

2.1. Functional Requirement #1: Background Crowd Data

2.1.1 #1-1 AggregateCrowdData

Use Case ID:	#1-1		
Use Case Name:	AggregateCrowdData		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	06/11/2026

Actor:	System
Description:	Periodically collects crowd data from external sources and merges it into one unified venue list.
Preconditions:	1. None. Runs on a schedule and whenever cached data expires.
Postconditions:	1. An up-to-date, merged venue list is available to serve users.
Priority:	High
Frequency of Use:	High. Every 5 minutes for stations, weekly per mall for Popular Times patterns.
Flow of Events:	1. System checks whether the cached station data has expired (5 minutes). 2. If it has, system invokes FetchTransitCrowdData (#1-2). 3. System checks for malls whose Popular Times pattern is more than a week old. 4. If there are any, system invokes ScrapePopularTimes (#1-3) for them one at a time. 5. System fetches carpark lot counts from LTA DataMall (cached 1 minute) and attaches them to the matching malls as a bonus figure. 6. System merges station and mall data into one list. 7. System invokes DeclutterOverlappingPins (#1-4) before serving the list. 8. System invokes SyncPopularTimesCache (#1-5) on start-up and every 15 minutes.
Alternative Flows:	1-1.AC.1 If a source fails, the system keeps serving its last known data: each mall's last pattern, however old, and each station's last reading for up to an hour. 1-1.AC.2 When the scraping backend starts, if some malls were scraped less than a week ago, it asks in its terminal whether to rescrape them too, accepting yes, y, no or n in any case. If the operator answers yes, ScrapePopularTimes (#1-3) runs once for every mall. If the answer is no, or there is no terminal to ask in, only malls more than a week old are scraped.
Exceptions:	1-1.EX.1 If a source fails for only some records, only those records are skipped.
Includes:	FetchTransitCrowdData (#1-2), ScrapePopularTimes (#1-3), DeclutterOverlappingPins (#1-4), SyncPopularTimesCache (#1-5)
Special Requirements:	Requires a valid LTA DataMall API account key.
Assumptions:	None

Notes and Issues:	Carpark lots are shown on a mall's page only. They are not used to estimate crowd levels.

2.1.2 #1-2 FetchTransitCrowdData

Use Case ID:	#1-2		
Use Case Name:	FetchTransitCrowdData		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	20/10/2026

Actor:	System
Description:	Fetches real-time MRT/LRT crowd density from LTA DataMall.
Preconditions:	1. Cached station data has expired.
Postconditions:	1. Current crowd levels for all tracked stations are retrieved.
Priority:	High
Frequency of Use:	High. Every 5 minutes.
Flow of Events:	1. System calls LTA DataMall's Station Crowd Density API. 2. System maps each station's raw crowd code to a Low, Moderate or High level. 3. System checks each station's real operating hours. A closed station is marked Closed regardless of the raw reading.
Alternative Flows:	None
Exceptions:	1-2.EX.1 If the API is unreachable, each station keeps its last reading for up to one hour, then shows Unavailable.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	LTA's feed has no notion of closed. It is derived separately from each station's real first and last train times.

2.1.3 #1-3 ScrapePopularTimes

Use Case ID:	#1-3		
Use Case Name:	ScrapePopularTimes		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	06/11/2026

Actor:	System
Description:	Retrieves a mall's Google Popular Times pattern and opening hours, since no official API exists for them.
Preconditions:	1. A mall's pattern is more than a week old, a user opened a mall whose pattern is more than a week old, a user pressed Refresh on a mall's page (RefreshVenueCrowdData (#3-4)), or the operator chose to rescrape every mall when the scraping backend started (AggregateCrowdData (#1-1)).
Postconditions:	1. The mall's weekly hourly pattern and opening hours are updated in the local cache and the shared database.
Priority:	High
Frequency of Use:	Weekly per mall. While malls are due, one mall after another, about 10 seconds each.

Flow of Events:	1. System opens Google Maps for the mall in a Chrome profile signed in to the project's Google account. 2. System opens the mall's place page. 3. System scrolls to load the Popular Times chart. 4. System extracts the full week's hourly pattern. 5. System reads the opening hours for each day. If they are folded behind "See more hours", system expands them first. 6. System caches the result and uploads it to the shared database.
Alternative Flows:	1-3.AC.1 If Google shows its signed-out "limited view" (no Popular Times), scraping pauses and the server log asks for the account to be signed in again. 1-3.AC.2 If Google lists no opening hours for the mall, known fixed hours are used where the team has them.
Exceptions:	1-3.EX.1 If the chart never loads, the mall keeps its last cached reading.
Includes:	None
Special Requirements:	Headless Chrome with a signed-in Google profile, on one designated machine.
Assumptions:	Popular Times data is collected one mall at a time, for a non-commercial academic project.
Notes and Issues:	Depends on an unofficial page layout, so it can break if Google changes the page. Since October 2026, Google only shows Popular Times to signed-in visitors.

2.1.4 #1-4 DeclutterOverlappingPins

Use Case ID:	#1-4		
Use Case Name:	DeclutterOverlappingPins		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	System
Description:	Repositions venues that share a near-identical map location so each remains individually visible and selectable.
Preconditions:	1. A merged list of venues with coordinates is available.
Postconditions:	1. No two venues occupy the same small map area.
Priority:	Medium
Frequency of Use:	High
Flow of Events:	1. System groups venues by rounding their coordinates to a small grid cell. 2. For each group with more than one venue, system computes the shared centre point. 3. System arranges the group's venues in a small circle (about 39 m) around that centre.
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	This replaces marker clustering, which Crowdy no longer uses. Every venue stays individually tappable at any zoom level.

2.1.5 #1-5 SyncPopularTimesCache

Use Case ID:	#1-5		
Use Case Name:	SyncPopularTimesCache		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	20/10/2026	Date Last Updated:	20/10/2026

Actor:	System
Description:	Keeps each backend's Popular Times cache in step with the team's shared copy in the database, so a pattern scraped on one machine is available to every backend.
Preconditions:	1. Database connection details are configured.
Postconditions:	1. The local cache and the shared database both hold the newest pattern for every mall.
Priority:	Medium
Frequency of Use:	On start-up, every 15 minutes, and before each scraping session.
Flow of Events:	1. System loads all stored patterns from the shared database. 2. For each mall whose shared pattern is newer than the local one, system copies it into the local cache. 3. For each mall whose local pattern is newer, system uploads it to the database. 4. Whenever a new pattern is scraped, system also uploads it immediately.
Alternative Flows:	1-5.AC.1 If the database is unreachable, system logs it once and keeps working from its local cache. The next sync retries.
Exceptions:	1-5.EX.1 A failed scrape (no pattern) never overwrites a stored pattern.
Includes:	None
Special Requirements:	MySQL database (hosted on Aiven) with its SSL certificate.
Assumptions:	Only one backend scrapes. The others only read from the database.
Notes and Issues:	None

2.2. Functional Requirement #2: Map and Nearby Venues

2.2.1 #2-1 ViewNearbyVenues

Use Case ID:	#2-1		
Use Case Name:	ViewNearbyVenues		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	08/11/2026

Actor:	User
Description:	Displays a map and a list of nearby venues (malls and MRT/LRT stations) with their current crowd levels.
Preconditions:	1. None. Location permission is optional.
Postconditions:	1. The map and list show the venues in view, each with a crowd level.
Priority:	High
Frequency of Use:	High

Flow of Events:	1. User opens the app. 2. System requests the user's current location. 3. System retrieves the venue list via AggregateCrowdData (#1-1). 4. System shows the whole of Singapore with the crowd heatmap on, and the crowd snapshot card, ViewCrowdSnapshot (#2-4), above the list. 5. User turns on the Venue or Transit layer (ToggleMapLayer (#2-2)), and the system displays those venues as crowd-coloured dots on the map. 6. System displays the same venues as a list of cards below the map, sorted by distance from the user. 7. As the user pans or zooms, the list updates to match the visible map area. 8. User taps a venue's card or dot. 9. System expands that card in place with its crowd line and, for a mall, a See details button. It fades the other dots on the map and draws the selected dot on top. 10. While the venue is selected, the list holds still even as the map moves. When the user deselects it, the list catches up with the map.
Alternative Flows:	2-1.AC.1 If location permission is denied, the cards show that distance is unavailable and the list is not sorted by distance. 2-1.AC.2 If the user taps a dot on the map, the list scrolls once to that venue's card. 2-1.AC.3 If the user taps See details on a mall's card, ViewVenueDetail (#3-1) is invoked.
Exceptions:	2-1.EX.1 If no venues are available in the visible area, an empty-state message is shown. 2-1.EX.2 If both layers are off, the list asks the user to toggle on a venue type.
Includes:	AggregateCrowdData (#1-1), ViewCrowdSnapshot (#2-4)
Special Requirements:	Device location services (GPS).
Assumptions:	None
Notes and Issues:	A mall's expanded card reads, for example, "41% of peak · updated as of 4:00 pm" (the current hour), or "0% of peak · Opens 10 am" when closed. A station's card shows whether it is operating, its next first or last train time, and when its reading was updated. The venue list refreshes every 2 minutes while the app is open.

2.2.2 #2-2 ToggleMapLayer

Use Case ID:	#2-2		
Use Case Name:	ToggleMapLayer		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	08/11/2026

Actor:	User
Description:	Shows or hides Transit or Venue dots on the map and the matching cards in the list.
Preconditions:	1. None
Postconditions:	1. The selected layer's dots and list cards appear or disappear.
Priority:	Medium

Frequency of Use:	High
Flow of Events:	1. User taps the Transit or Venue layer button on the map. 2. System shows or hides that category's dots and list cards.
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Both layers are off by default, so the app opens on the crowd heatmap. Until a layer is on, the list asks the user to toggle on a venue type.

2.2.3 #2-3 RecentreMap

Use Case ID:	#2-3		
Use Case Name:	RecentreMap		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Recentres the map on the user's current location.
Preconditions:	1. User has moved the map away from their location.
Postconditions:	1. The map recentres on the user's current GPS location.
Priority:	Low
Frequency of Use:	Medium
Flow of Events:	1. User taps the recentre button. 2. System retrieves the current GPS location. 3. Map animates back to centre on it.
Alternative Flows:	None
Exceptions:	2-3.EX.1 If the location is unavailable, an alert is shown.
Includes:	None
Special Requirements:	Device location services (GPS).
Assumptions:	None
Notes and Issues:	None

2.2.4 #2-4 ViewCrowdSnapshot

Use Case ID:	#2-4		
Use Case Name:	ViewCrowdSnapshot		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	20/10/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Shows an at-a-glance summary of crowds across Singapore at the top of the Areas list.
Preconditions:	1. The venue list has loaded.
Postconditions:	1. The card shows mall counts by crowd level, the crowd level near the user, and the quietest open mall nearby.

Priority:	Medium
Frequency of Use:	High. Every time the app is opened.
Flow of Events:	1. User opens the app on the Areas tab. 2. System counts the malls that are quiet, moderate, busy and closed. 3. System finds the crowd level that most open malls and stations within 2 km share. 4. System finds the quietest open mall within 5 km. 5. System displays the card above the venue list. 6. User taps the quietest mall, invoking ViewVenueDetail (#3-1).
Alternative Flows:	2-4.AC.1 If the location is unavailable, the "near you" line is hidden. 2-4.AC.2 If no mall within 5 km is open, the quietest open mall anywhere is shown instead. 2-4.AC.3 If every mall is closed, the card says so instead.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	The "near you" line shows a crowd level, not an average percentage, because each mall's percentage is relative to its own peak and cannot be meaningfully averaged.

2.3. Functional Requirement #3: Venue Detail

2.3.1 #3-1 ViewVenueDetail

Use Case ID:	#3-1		
Use Case Name:	ViewVenueDetail		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Displays the full crowd detail for a mall.
Preconditions:	1. User tapped See details on a mall's card, the quietest mall on the snapshot card, an alternative, or a suggested meetup mall.
Postconditions:	1. The mall's crowd level, typical week, opening hours and nearby alternatives are displayed.
Priority:	High
Frequency of Use:	High

Flow of Events:	1. User opens a mall's page. 2. System retrieves the mall's full detail. 3. System displays its name, address, crowd level, crowd percentage and the line "Typical at this hour · source: Google Popular Times". 4. System displays the opening hours, with whether the mall is open now. The user can expand them to see the whole week. 5. System displays the typical crowd chart for today. The user can tap any hour to see its usual busyness. 6. System displays carpark lot availability where LTA has data for the mall. 7. System invokes ViewAlternatives (#3-2) to show quieter malls nearby.
Alternative Flows:	3-1.AC.1 If the user taps Directions, GetDirections (#3-3) is invoked. 3-1.AC.2 If the user taps Refresh, RefreshVenueCrowdData (#3-4) is invoked. 3-1.AC.3 If the user taps the heart, ManageFavourites (#6-4) is invoked.
Exceptions:	3-1.EX.1 If the mall's data is no longer available, a "not found" message is shown.
Includes:	ViewAlternatives (#3-2). Extended by GetDirections (#3-3) and RefreshVenueCrowdData (#3-4).
Special Requirements:	None
Assumptions:	None
Notes and Issues:	The crowd level shows Closed whenever the mall's opening hours say it is closed. On opening, the system refreshes Popular Times for this mall only if its pattern is more than a week old. MRT/LRT stations have no separate page. Their details are shown in their expanded list card instead.

2.3.2 #3-2 ViewAlternatives

Use Case ID:	#3-2		
Use Case Name:	ViewAlternatives		
Created By:	Xu Hanwen	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	21/10/2026

Actor:	User
Description:	Recommends lower-crowd venues of the same category near the currently viewed venue.
Preconditions:	1. User is viewing a venue's detail screen.
Postconditions:	1. Lower-crowd alternatives, if any, are displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. System identifies venues of the same category. 2. System filters to those within 3 km. 3. System filters to those with a lower crowd percentage. 4. System sorts the remaining venues by distance. 5. System displays up to three results.
Alternative Flows:	3-2.AC.1 If the user selects an alternative, ViewVenueDetail (#3-1) is invoked for it.
Exceptions:	3-2.EX.1 If no qualifying alternative exists, nothing is shown in this section.

Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Transit venues are excluded from alternatives matching.

2.3.3 #3-3 GetDirections

Use Case ID:	#3-3		
Use Case Name:	GetDirections		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Teo Kok Wei, Oscar
Date Created:	20/10/2026	Date Last Updated:	20/10/2026

Actor:	User
Description:	Shows a route from the user's location to the venue on Crowdy's own map, by public transport, walking, driving or cycling, with the live crowd level at each MRT/LRT station boarded.
Preconditions:	1. User is viewing a venue's detail, and location permission is granted.
Postconditions:	1. The route is drawn on the map with step-by-step instructions.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User taps Directions on the venue's detail. 2. System gets the user's current location. 3. System requests a route for the selected mode (public transport by default) from OneMap. 4. System draws the route on the map, coloured by MRT line, and lists the steps. 5. For each MRT/LRT leg, system shows the boarding station's current crowd level. 6. User switches to another mode, and steps 3 to 5 repeat. 7. If the user moves more than 150 m, system recalculates the route.
Alternative Flows:	3-3.AC.1 If location permission is denied, a message explains that location is needed.
Exceptions:	3-3.EX.1 If OneMap is unavailable, system shows "Couldn't load this route right now" with a Try again button.
Includes:	None
Special Requirements:	OneMap account credentials on the server. Device location services.
Assumptions:	None
Notes and Issues:	Routes are computed by OneMap. A route can occasionally loop round a block near its start or end, because OneMap starts from the nearest road or path it knows.

2.3.4 #3-4 RefreshVenueCrowdData

Use Case ID:	#3-4		
Use Case Name:	RefreshVenueCrowdData		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	27/10/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Lets the user ask for a fresh Popular Times reading for the mall they are viewing.
Preconditions:	1. User is viewing a mall's detail page.
Postconditions:	1. The page shows the newest available reading, with a short message saying what the refresh did.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. User taps Refresh on the mall's page. 2. System checks when the mall was last refreshed. 3. If it was not refreshed in the last 15 minutes, system invokes ScrapePopularTimes (#1-3) for the mall. 4. System updates the page with the new reading and shows "Updated just now."
Alternative Flows:	3-4.AC.1 If the mall was refreshed in the last 15 minutes, system keeps the current reading and shows "Already up to date - it was refreshed in the last 15 minutes." 3-4.AC.2 If this backend is not the one that scrapes, system loads the newest copy from the shared database and shows "Showing the latest shared data."
Exceptions:	3-4.EX.1 If the scrape fails, system keeps the last reading and shows "Couldn't refresh right now - showing the last reading." 3-4.EX.2 If the same client refreshes too often, system shows "Too many refreshes - wait a minute and try again."
Includes:	ScrapePopularTimes (#1-3)
Special Requirements:	Manual refreshes are rate limited per client.
Assumptions:	None
Notes and Issues:	Malls only. Station readings already refresh every 5 minutes.

2.4. Functional Requirement #4: Crowd Heatmap

2.4.1 #4-1 ViewCrowdHeatmap

Use Case ID:	#4-1		
Use Case Name:	ViewCrowdHeatmap		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	08/11/2026

Actor:	User
Description:	Displays an overlay where busy malls and MRT/LRT stations (Moderate and above) glow yellow to red, so hotspots stand out.

Preconditions:	1. User is viewing the map.
Postconditions:	1. The heatmap overlay is shown or hidden on the map.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User taps the heatmap toggle. 2. System invokes GenerateHeatmapImage (#4-2) if no fresh image is cached. 3. System places the image on the map, positioned to its real geographic bounds. 4. User taps the toggle again to hide it.
Alternative Flows:	None
Exceptions:	4-1.EX.1 If image generation fails, the toggle has no visible effect.
Includes:	GenerateHeatmapImage (#4-2)
Special Requirements:	iOS only. react-native-maps has no gradient or heatmap layer outside Google Maps, which Expo Go no longer supports.
Assumptions:	None
Notes and Issues:	Quiet, closed and no-data venues contribute no colour. The heatmap is on by default, so it shows as soon as the app opens.

2.4.2 #4-2 GenerateHeatmapImage

Use Case ID:	#4-2		
Use Case Name:	GenerateHeatmapImage		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	20/10/2026

Actor:	System
Description:	Renders the crowd heatmap as a single blurred, colour-graded image.
Preconditions:	1. ViewCrowdHeatmap (#4-1) was invoked and no fresh cached image exists.
Postconditions:	1. A PNG image representing current crowd levels across all open venues is available.
Priority:	Medium
Frequency of Use:	On demand, cached for about 5 minutes.
Flow of Events:	1. System filters to malls and stations at Moderate crowd level or above. 2. System projects each venue's coordinates onto a canvas. 3. System draws a soft radial colour gradient per venue on a transparent background. 4. System exports the canvas as a PNG.
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	Rendered in a headless browser on the server, separate from the signed-in browser used by ScrapePopularTimes (#1-3).
Assumptions:	None
Notes and Issues:	Stations have no percentage, so each is coloured at the middle of its level's colour band to match its badge.

2.5. Functional Requirement #5: Search

2.5.1 #5-1 SearchVenue

Use Case ID:	#5-1		
Use Case Name:	SearchVenue		
Created By:	K Eswaran	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Lets the user search the full tracked venue list by name, address or postal code.
Preconditions:	1. User is on the Search tab.
Postconditions:	1. Matching venues are displayed, best match first.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. User types a venue name, address or postal code. 2. System matches names, address words ("rd" = road) and postal codes, at the start of words only, forgiving one-letter typos in words (not in postal codes). 3. System ranks the results: exact postal code, postal code prefix, name prefix, name words, address words, then typo matches. 4. User taps a result to expand its card. 5. For a mall, the user can tap See details, invoking ViewVenueDetail (#3-1).
Alternative Flows:	None
Exceptions:	5-1.EX.1 If no venues match, a "no results" message is shown.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Searching runs in the app against the already-loaded venue list, with no separate search request. Address, postal code and typo matching were added by Mao Yikai.

2.6. Functional Requirement #6: Accounts and Saved Venues

2.6.1 #6-1 AccountAuthentication

Use Case ID:	#6-1		
Use Case Name:	AccountAuthentication		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Lets the user sign up for a new account or log in to an existing one.
Preconditions:	1. User is not signed in.
Postconditions:	1. User is authenticated and holds a session token.
Priority:	Medium

Frequency of Use:	Low
Flow of Events:	1. User opens Settings and selects Sign Up or Log In. 2. User enters a username and password. 3. System validates the credentials. 4. If the user had saved venues on the phone as a guest, ManageFavourites (#6-4) is invoked to offer adding them to the account. 5. System returns the user to Settings, now signed in. 6. System invokes NotifyFriendRequests (#8-5) to tell the user about any friend requests waiting for them.
Alternative Flows:	None
Exceptions:	6-1.EX.1 A wrong username or password shows the same generic error either way, so login attempts cannot be used to check which usernames exist.
Includes:	LoginAccount (#6-2), CreateAccount (#6-3), NotifyFriendRequests (#8-5). Extended by ManageFavourites (#6-4).
Special Requirements:	Passwords are hashed with bcrypt. Sessions are signed tokens that expire after 30 days. Sign-in attempts are rate limited per client.
Assumptions:	None
Notes and Issues:	Accounts use a username, not an email address.

2.6.2 #6-2 LoginAccount

Use Case ID:	#6-2		
Use Case Name:	LoginAccount		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Allows a user to log in to an existing account.
Preconditions:	1. User has an existing account.
Postconditions:	1. User is authenticated and returned to Settings.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. User enters their username and password. 2. System validates the credentials against the database. 3. If they are correct, system issues a session token.
Alternative Flows:	None
Exceptions:	6-2.EX.1 If the credentials are incorrect, a generic error is shown. 6-2.EX.2 If the same client tries too often, system shows "Too many attempts - wait a minute and try again."
Includes:	None
Special Requirements:	None
Assumptions:	User has already registered.
Notes and Issues:	None

2.6.3 #6-3 CreateAccount

Use Case ID:	#6-3		
Use Case Name:	CreateAccount		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Allows a visitor to create a new account.
Preconditions:	1. None
Postconditions:	1. A new account is created and stored in the database.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. User enters a username and password. 2. System validates the username format (3 to 24 letters, numbers or underscores) and the password length (at least 8 characters). 3. System checks that the username is not already taken. 4. System hashes the password and stores the new account. 5. System issues a session token.
Alternative Flows:	None
Exceptions:	6-3.EX.1 If the username is already taken, or the input is invalid, an error is shown. 6-3.EX.2 If the same client tries too often, system shows "Too many attempts - wait a minute and try again."
Includes:	None
Special Requirements:	The password is stored only as a bcrypt hash, never in plain text.
Assumptions:	None
Notes and Issues:	None

2.6.4 #6-4 ManageFavourites

Use Case ID:	#6-4		
Use Case Name:	ManageFavourites		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Mao Yikai
Date Created:	16/09/2026	Date Last Updated:	20/10/2026

Actor:	User
Description:	Save or remove venues as favourites, and view them in a dedicated list.
Preconditions:	1. None
Postconditions:	1. The venue is added to or removed from favourites.
Priority:	Medium
Frequency of Use:	Medium

Flow of Events:	1. User taps the heart on a venue card or on the venue's detail screen. 2. If the user is signed in, system saves or removes the favourite against the account. 3. If not, system saves or removes it in local on-device storage. 4. If the user later signs in with local favourites present, they are explicitly asked (not silently merged) whether to bring them into the account. The on-device saves are only cleared after the account update succeeds. If it fails, they are kept and the user is asked again at the next sign-in.
Alternative Flows:	None
Exceptions:	6-4.EX.1 If a favourited venue no longer exists in the data source, it is silently dropped from the list.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Extends AccountAuthentication (#6-1) when the phone holds guest saves at sign-in. The explicit merge prompt stops one guest's leftover local saves from being added to every account later signed in on the same device.

2.6.5 #6-5 ViewSavedVenues

Use Case ID:	#6-5		
Use Case Name:	ViewSavedVenues		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Joseph Tan
Date Created:	16/09/2026	Date Last Updated:	16/09/2026

Actor:	User
Description:	Displays all venues the user has favourited.
Preconditions:	1. None
Postconditions:	1. A list of saved venues, each with its current crowd level, is displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User selects the Saved tab. 2. System retrieves the user's favourited venue ids (from the account or the phone, depending on sign-in state). 3. System displays them sorted by distance.
Alternative Flows:	None
Exceptions:	6-5.EX.1 If no venues are saved, an empty-state message is shown.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

2.6.6 #6-6 LogOut

Use Case ID:	#6-6		
Use Case Name:	LogOut		
Created By:	Teo Kok Wei, Oscar	Last Updated By:	Teo Kok Wei, Oscar
Date Created:	16/09/2026	Date Last Updated:	16/09/2026

Actor:	Registered User
Description:	Ends the current session.
Preconditions:	1. User is signed in.
Postconditions:	1. The session token is discarded, and the user reverts to the guest, on-device-only state.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. User selects Log Out in Settings. 2. System discards the stored session token. 3. Saved venues are now read from the phone instead of the account.
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Also triggered automatically if the backend ever rejects the session token as expired or invalid.

2.7. Functional Requirement #7: Preferences

2.7.1 #7-1 ChangeLanguage

Use Case ID:	#7-1		
Use Case Name:	ChangeLanguage		
Created By:	Mao Yikai	Last Updated By:	Mao Yikai
Date Created:	20/10/2026	Date Last Updated:	20/10/2026

Actor:	User
Description:	Switches the app's text between English and Simplified Chinese.
Preconditions:	1. None
Postconditions:	1. All app text is shown in the chosen language, and the choice is remembered.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. User opens Settings. 2. User selects English or 中文. 3. System updates every screen immediately. 4. System saves the choice on the device.
Alternative Flows:	None
Exceptions:	None

Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Venue, street and station names stay in English, since the source data only has English names. OneMap's turn-by-turn steps are translated.

2.7.2 #7-2 ChangeTheme

Use Case ID:	#7-2		
Use Case Name:	ChangeTheme		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	27/08/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Switches the app between dark mode and light mode.
Preconditions:	1. None
Postconditions:	1. Every screen, including the map, uses the chosen theme, and the choice is remembered.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. User opens Settings. 2. User switches Dark Mode on or off. 3. System restyles every screen, the map and the status bar immediately. 4. System saves the choice on the device.
Alternative Flows:	None
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	Dark mode is the default.

2.8. Functional Requirement #8: Friends

2.8.1 #8-1 SendFriendRequest

Use Case ID:	#8-1		
Use Case Name:	SendFriendRequest		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	22/10/2026	Date Last Updated:	01/11/2026

Actor:	Registered User
Description:	Sends a friend request to another Crowdy account by username.
Preconditions:	1. User is signed in.
Postconditions:	1. A pending request is stored, or the two accounts become friends if the other person had already sent a request.
Priority:	Medium

Frequency of Use:	Low
Flow of Events:	1. User opens Friends, from Settings or from Manage friends on the Meet up tab. 2. User enters the other person's username under Add a Friend and taps Add. 3. System checks the username and stores a pending request. 4. System shows "Request sent to {name}." and lists the request under Sent. 5. The other person is told about it through NotifyFriendRequests (#8-5).
Alternative Flows:	8-1.AC.1 If the other person had already sent the user a request, the two accounts become friends at once and system shows "You and {name} are now friends." 8-1.AC.2 The user can tap Cancel on a sent request to withdraw it.
Exceptions:	8-1.EX.1 If the username cannot be used, system shows "Couldn't send a friend request to that username." without saying whether the account exists. 8-1.EX.2 If the user enters their own username, system shows "You can't add yourself." 8-1.EX.3 If they are already friends, system shows "You're already friends." 8-1.EX.4 If a request is already pending, system shows "You've already sent them a request." 8-1.EX.5 If the same client sends too many requests, system shows "Too many friend requests - wait a minute and try again."
Includes:	None
Special Requirements:	Friend requests are rate limited per client.
Assumptions:	None
Notes and Issues:	None

2.8.2 #8-2 RespondToFriendRequest

Use Case ID:	#8-2		
Use Case Name:	RespondToFriendRequest		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	22/10/2026	Date Last Updated:	01/11/2026

Actor:	Registered User
Description:	Accepts or declines a friend request the user has received.
Preconditions:	1. User is signed in and has at least one incoming request.
Postconditions:	1. If accepted, the two accounts are friends and each can see the other's home station. If declined, the request is removed.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. User opens Friends, or taps View on a friend request alert. 2. System lists the incoming requests under Requests. 3. User taps Accept. 4. System marks the request accepted. 5. System moves the person to the Friends list with their home station, and clears the request count on the Settings tab and Friends row.

Alternative Flows:	8-2.AC.1 If the user taps Decline, system removes the request. The sender is not told.
Exceptions:	8-2.EX.1 If the request no longer exists (the sender cancelled it), an error message is shown.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

2.8.3 #8-3 RemoveFriend

Use Case ID:	#8-3		
Use Case Name:	RemoveFriend		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	22/10/2026	Date Last Updated:	01/11/2026

Actor:	Registered User
Description:	Removes an accepted friend.
Preconditions:	1. User is signed in and has at least one friend.
Postconditions:	1. The friendship is deleted. Neither person can see the other's home station or add the other from their friends list.
Priority:	Low
Frequency of Use:	Low
Flow of Events:	1. User opens Friends. 2. User taps Remove next to a friend. 3. System asks "Remove {name} from your friends?" 4. User confirms. 5. System deletes the friendship and updates the list.
Alternative Flows:	8-3.AC.1 If the user cancels the confirmation, nothing changes.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

2.8.4 #8-4 SetHomeStation

Use Case ID:	#8-4		
Use Case Name:	SetHomeStation		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	22/10/2026	Date Last Updated:	01/11/2026

Actor:	Registered User
Description:	Sets the MRT/LRT station the user usually starts from, so friends can add them to a meetup without typing a station.
Preconditions:	1. User is signed in.
Postconditions:	1. The home station is saved. Friends who add the user to a meetup start them from it.

Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. User opens Friends. 2. User taps Change under Your Home Station. 3. User searches for and picks a station in the station picker. 4. System saves the station and shows its name.
Alternative Flows:	8-4.AC.1 If the user taps Clear, system removes the home station. A friend planning a meetup then picks a station for them.
Exceptions:	None
Includes:	None
Special Requirements:	A home station is only visible to accepted friends.
Assumptions:	None
Notes and Issues:	None

2.8.5 #8-5 NotifyFriendRequests

Use Case ID:	#8-5		
Use Case Name:	NotifyFriendRequests		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	27/10/2026	Date Last Updated:	01/11/2026

Actor:	Registered User
Description:	Tells a signed-in user about friend requests waiting for them.
Preconditions:	1. User is signed in.
Postconditions:	1. The user has been alerted to each new request once, and the waiting requests are counted on the Settings tab and Friends row.
Priority:	Low
Frequency of Use:	Every sign-in, app start and return to the app.
Flow of Events:	1. User signs in, opens the app while signed in, or returns to the app. 2. System fetches the user's incoming friend requests. 3. For requests not yet shown since sign-in, system shows an alert: "{name} sent you a friend request." (or "You have {count} new friend requests.") with Later and View. 4. System shows a red count on the Settings tab and on the Friends row in Settings. 5. If the user taps View, system opens Friends, where RespondToFriendRequest (#8-2) can be performed.
Alternative Flows:	8-5.AC.1 If the user taps Later, the alert closes and the counts remain. 8-5.AC.2 If there are no new requests, no alert is shown.
Exceptions:	8-5.EX.1 If the server cannot be reached, no alert is shown. System checks again the next time the user returns to the app.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	In-app only. Crowdy does not send phone push notifications.

2.9. Functional Requirement #9: Meetup Planner

2.9.1 #9-1 PlanMeetup

Use Case ID:	#9-1		
Use Case Name:	PlanMeetup		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	22/10/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Suggests malls where a group can meet that are fair for everyone to reach by public transport and not too crowded.
Preconditions:	1. None. The planner works without an account. Adding friends needs one.
Postconditions:	1. Up to three suggested malls are shown, each with every person's travel time and the mall's crowd level.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. User opens the Meet up tab. 2. User adds 2 to 6 people: Add me (their own home station), Add a friend (the friend's home station) or Add someone (any station). 3. User picks what matters most: Fairest for everyone (default), Shortest overall or Least crowded. 4. User picks when: Now, or a day up to a week ahead with an hour between 7 am and 11 pm (7 pm by default). 5. User taps Find places to meet. 6. System shortlists malls by straight-line distance, then asks OneMap for each person's public transport time to each shortlisted mall. 7. System ranks the malls by the chosen priority, with busier malls penalised. Least crowded only considers malls where no one's trip is more than 20 minutes longer than the best option. 8. System shows the top three malls with the longest and average trip, each person's time and the mall's crowd level. 9. User taps See everyone's routes, invoking ViewGroupRoutes (#9-2), or taps the mall, invoking ViewVenueDetail (#3-1).
Alternative Flows:	9-1.AC.1 For a later time, system uses each mall's usual crowd at that hour and the train and bus timetable then. 9-1.AC.2 If a friend has no home station, the user picks a station for them before searching. 9-1.AC.3 If every mall is closed at the chosen time, system says so. 9-1.AC.4 When the group reaches 6 people, the add buttons are replaced by "Maximum group size reached (max 6)." 9-1.AC.5 If the user is not signed in, Add a friend asks them to sign in from Settings.

Exceptions:	9-1.EX.1 If fewer than two people have a starting station, Find places to meet is disabled with "Add at least two people, each with a starting station." 9-1.EX.2 If OneMap cannot plan a trip, its time is estimated and marked ≈. 9-1.EX.3 If the chosen time has passed, system shows "That time has passed - choose another." 9-1.EX.4 If the same client searches too often, system shows "Too many searches - wait a minute and try again." 9-1.EX.5 If the server cannot be reached, system shows "Couldn't get suggestions right now."
Includes:	None. Extended by ViewGroupRoutes (#9-2).
Special Requirements:	OneMap account credentials on the server. Meetup searches are rate limited per client. At most 8 route lookups run at once, and repeated trips are cached for 6 hours.
Assumptions:	Each person starts from an MRT/LRT station.
Notes and Issues:	Several people may start from the same station.

2.9.2 #9-2 ViewGroupRoutes

Use Case ID:	#9-2		
Use Case Name:	ViewGroupRoutes		
Created By:	Joseph Tan	Last Updated By:	Joseph Tan
Date Created:	27/10/2026	Date Last Updated:	01/11/2026

Actor:	User
Description:	Shows everyone's public transport route to one suggested mall on a single map.
Preconditions:	1. PlanMeetup (#9-1) has returned suggestions.
Postconditions:	1. Each person's route, leave time and arrive time are shown.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. User taps See everyone's routes on a suggestion. 2. System asks OneMap for each person's public transport route to the mall. 3. System draws every route on one map in that person's colour, with numbered start markers and the mall. 4. System lists each person's leave and arrive times. 5. User taps a person. 6. System highlights that person's route in its real MRT, bus and walking colours and shows their steps.
Alternative Flows:	9-2.AC.1 If a meeting time was chosen, the user picks Arrive by (default, each person leaves early enough) or Leave at (everyone sets off then), and the routes are planned again. 9-2.AC.2 The user can drag the list up over the map for long itineraries.
Exceptions:	9-2.EX.1 If OneMap cannot plan a trip, its time is shown as an estimate. 9-2.EX.2 If the routes cannot be loaded, system shows "Couldn't get the routes right now."
Includes:	None

Special Requirements:	OneMap account credentials on the server.
Assumptions:	None
Notes and Issues:	Live station crowd levels appear on the steps only when meeting now, since a live reading would be misleading for a later trip.